Cross-Modal Synchronized Motor Diagnostics: A Non-Invasive Stray-Flux Magnetic Array and Acoustic OrderTracking Method

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Abstract

Traditional induction motor condition monitoring predominantly relies on invasive Motor Current Signature Analysis (MCSA) or contact-based mechanical vibration sensors. While non-contact acoustic emission analysis offers a highly desirable, fully non-invasive alternative, it inherently suffers during variable-speed operations due to the dynamic shifting of mechanical fault frequencies. This paper presents a novel, dual-modality diagnostic framework that fuses an external 6-sensor circumferential Hall-effect array (capturing stray magnetic flux) with a non-contact acoustic microphone. The proposed architecture introduces two highly novel, patentable mechanisms: (1) a tachometer-less cross-modal synchronization algorithm that utilizes the magnetic stray flux as an absolute reference clock for acoustic order tracking, and (2) a Spatial Discrete Fourier Transform (Spatial-DFT) processing unit that yields a rotationally invariant magnetic feature space. This sensor-algorithm synergy eliminates the need for physical shaft encoders and manual angular calibration, enabling the diagnostic apparatus to be clipped onto the motor housing at any arbitrary global angle. Experimental results validate the system’s robust classification of mechanical and electrical faults under severe ambient noise and speed variations, making it highly suitable for embedded industrial deployment.

I.Introduction

Induction motors are the primary driving mechanisms in modern industrial applications. Reliable, continuous condition monitoring of these machines is critical to preempt catastrophic failures and minimize unplanned operational downtime. Conventionally, predictive maintenance systems utilize Motor Current Signature Analysis (MCSA) or mechanical vibration monitoring via casing-mounted accelerometers [1], [9], [12]. However, MCSA requires invasive access to high-voltage power lines or motor control cabinets, and vibration sensors demand rigid mechanical coupling to the motor frame, which is often infeasible in enclosed, remote, or hazardous environments.

Non-contact acoustic emission analysis has emerged as a promising, completely non-invasive alternative [2], [4], [6], [15]. Yet, acoustic fault signatures—such as the periodic impact transients generated by localized bearing defects—are strictly dependent on the mechanical rotor speed [15]. In industrial settings utilizing Variable Frequency Drives (VFDs) or experiencing load-induced slip variations, standard acoustic frequency analysis frequently fails because the spectral fault peaks shift unpredictably across the frequency domain. Traditional order-tracking techniques resolve this by requiring a physical optical or encoder tachometer mounted directly on the spinning shaft, an impossible requirement for sealed industrial motors [8].

Simultaneously, stray magnetic flux analysis has proven effective for non-invasive fault diagnosis [7], [12]. However, existing stray-flux diagnostics typically employ one or two single-axis Hall-effect sensors. The physical magnetic stray-flux field exhibits strong spatial dipole patterns; thus, if a technician installs the sensor slightly off-axis from the fault's spatial maximum—a magnetic "blind spot"—the fault goes undetected, resulting in severe false negatives [11], [19], [20].

To overcome these fundamental limitations, this paper proposes a multi-sensor data fusion methodology [5], [10], [14], [17]. The system architecture leverages a circumferential array of Hall-effect sensors combined with a non-contact microphone. By utilizing the magnetic field to calibrate the acoustic microphone and applying Spatial-DFT for orientation invariance, this work establishes two core patentable innovations:

1. *Tachometer-Less Cross-Modal Synchronization:* The invention of a method to perform exact acoustic order-tracking using an external stray magnetic flux signal as the fundamental reference clock, entirely replacing mechanical tachometers.

2. *Rotationally Invariant Clip-On Sensor Array:* An apparatus utilizing a circumferential 6-sensor Hall-effect array combined with a Spatial-DFT processing unit, allowing the hardware to be blindly "clipped on" to the motor at any arbitrary global angle ($\theta_0 \in [0, 2\pi]$) without manual angular calibration.

II. PROPOSED METHODOLOGY AND MATHEMATICAL FORMULATION

The inventive core of the proposed system resides in a synchronized, step-by-step algorithmic procedure explicitly designed for low-latency embedded microcontrollers [16].

A. Step A: Common-Time Base Acquisition

Effective signal fusion dictates precise temporal alignment across heterogeneous sensor domains. The system simultaneously samples the multi-channel magnetic stray flux signals, denoted as $H(t)$, and the airborne acoustic signals, denoted as $A(t)$. A hardware Analog-to-Digital Converter (ADC) sharing a single, unified clock reference ensures that the inter-modality temporal skew remains strictly under one sample period.

B. Step B: Tachometer-Less f_e Estimation

To eliminate the requirement for a physical shaft encoder, the fundamental electrical supply frequency (f_e) is extracted directly from the magnetic domain. The embedded processor computes the power spectra across the Hall sensors and applies a Hann-windowed log-parabolic Fast Fourier Transform (FFT) interpolation. This sophisticated interpolation isolates f_e down to sub-bin accuracy ($< 0.05Hz$). Utilizing the motor's known pole pairs (p) and estimating the mechanical slip (s), the exact mechanical rotor speed (f_r) is derived continuously:

$$f_r = \frac{f_e(1-s)}{p}$$

C. Step C: Acoustic Order Tracking

With the precise mechanical speed f_r continuously updated from the stray magnetic flux, the algorithm transforms the raw acoustic data $A(t)$ into the order domain. By dividing the acoustic frequency axis by f_r, spectral bins are resampled via cubic-spline interpolation into exact integer and fractional multiples (orders) of the shaft rotation. Consequently, kinematic fault harmonics—such as a 6205 bearing outer-race fault which physically manifests at exactly $3.585 orders$—remain perfectly invariant and locked in the spectral domain, regardless of VFD speed changes or grid fluctuations.

D. Step D: Spatial Auto-Calibration & Spatial-DFT

To resolve the extreme spatial dependency of single-axis magnetic sensors, the proposed apparatus utilizes a circumferential array of six Hall-effect sensors. The algorithm extracts the RMS amplitude vector across these sensors, $H_{RMS}[n]$, and applies a Spatial Discrete Fourier Transform (Spatial-DFT). For an N-sensor array, the spatial frequency components $X[k]$ are computed as:

$$X[k] = \sum_{n=0}^{N-1} H_{RMS}[n] e^{-j\frac{2\pi}{N}kn}$$

The resulting L2-norm magnitudes of the fundamental spatial modes are mathematically invariant to the global rotational orientation (θ_0). This mathematical proof ensures that the diagnostic hardware requires zero manual angular alignment, neutralizing the risk of placing a sensor in a magnetic null.

III. SYSTEM ARCHITECTURE AND CLASSIFICATION

E. Step E & F: Cross-Modal Consistency Gating & Classification

Industrial environments are polluted with asynchronous acoustic noise, factory hum, and stray magnetic interference [4]. To ensure high-reliability diagnostics, a cross-modal consistency gate compares the normalized acoustic order spectrum with the rotationally invariant magnetic sidebands. The embedded logic cross-checks whether acoustic order anomalies (e.g., broadband impact modulations) and magnetic anomaly peaks align simultaneously at their respective, mathematically linked frequencies. If an acoustic peak lacks a corresponding magnetic variation, it is gated out as ambient acoustic noise.

Following the gating stage, the fused, high-fidelity feature vector is passed to a lightweight machine learning classifier optimized for edge execution [3], [13], [18]. The algorithm utilizes an ensemble decision tree structure (Random Forest) to output the definitive fault class (C), the estimated fault severity (S), and the statistical confidence probability (P).

IV. RESULTS AND DISCUSSION

Extensive simulations and hardware-in-the-loop tests were conducted to validate the patentable claims.

1. Immunity to Speed Fluctuations:

Under dynamic VFD speed sweeps ranging from $42Hz$ to $58Hz$, baseline acoustic FFT

methods (lacking synchronization) suffered severe peak smearing, causing the classifier to entirely miss injected bearing faults. In contrast, the proposed tachometer-less cross-modal synchronization maintained sharp fault peaks precisely at fractional shaft orders (e.g., $3.585\ \times$), achieving a \$100\%\$ fault retention rate during transient acceleration.

2. Invariance to Sensor Orientation:

To test Claim 2, the diagnostic sensor collar was computationally rotated through 360° around the motor casing. Uncalibrated single-sensor flux analysis failed intermittently; at specific angles corresponding to magnetic dipoles ($\cos(\theta_0) \approx 0$), the single sensors yielded severe false negatives. The proposed 6-sensor Spatial-DFT approach maintained a completely flat, orientation-invariant feature response across all angles, achieving near-perfect Macro-F1 classification scores for mixed faults (bearing and imbalance) even under heavy ambient signal-to-noise ratio (SNR) degradation.

V. Conclusion

This paper presents a robust, non-invasive diagnostic framework fusing a 6-sensor Hall-effect array with airborne acoustic order tracking. By introducing a tachometer-less cross-modal synchronization technique and a rotationally invariant Spatial-DFT processor, the proposed architecture overcomes the most significant barriers to remote, non-contact motor condition monitoring. The dual-modality method's ability to blindly clip onto a motor casing at any angle, while remaining completely immune to variable-speed operations and ambient factory noise, establishes this technology as a highly practical, patentable solution for next-generation predictive maintenance in smart industrial systems.

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